

# Declaring And Initializing Arrays

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## What We'll Learn

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By the end of this lesson, students can declare and initialize arrays in Java.

## Before We Start

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Imagine storing the scores of five students. Would you rather create five separate variables or store all the scores in one variable?

## Let's Understand

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An **array** is a variable that can store multiple values of the same data type.

Instead of creating several variables:

```
int score1 = 85;
int score2 = 90;
int score3 = 78;
```

We can store the values in one array:

```
int[] scores = {85, 90, 78};
```

All values inside an array must use the same data type.

### Declaring an Array

To declare an array, write the data type followed by square brackets `[]` and the array name.

```
int[] scores;
```

This declares an array variable named `scores`, but it does not yet create the array.

Other examples:

```
String[] names;
double[] prices;
boolean[] answers;
```

### Creating an Array

Use the `new` keyword to create an array with a specific number of elements.

```
scores = new int[5];
```

This creates an integer array that can store five values.

The declaration and creation can also be written in one statement:

```
int[] scores = new int[5];
```

The number inside the brackets determines the array's size.

```
String[] names = new String[3];
double[] prices = new double[4];
```

An array has a fixed size after it is created.

## Initializing an Array With Values

An array can be created and given values immediately.

```
int[] scores = {85, 90, 78, 92, 88};
```

Java automatically determines the size of the array based on the number of values provided.

The array above contains five integer values.

Other examples:

```
String[] subjects = {"Java", "HTML", "CSS"};
```

```
double[] prices = {25.50, 40.00, 75.25};
```

```
boolean[] answers = {true, false, true};
```

## Default Array Values

When an array is created using new, Java automatically gives each element a default value.

```
int[] numbers = new int[3];
```

The default values depend on the data type:

- `int` → 0
- `double` → 0.0
- `boolean` → false
- `String` → null

# Let's Practice

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## Practice 1

Declare an integer array named `studentScores` that can store five values.

## Practice 2

Declare and initialize a `String` array named `studentNames` with the following names:

- Angela
- Joshua
- Maria

## Practice 3

Declare and initialize a double array named `productPrices` with the following values:

- 25.50
- 50.00
- 75.25
- 100.00

## Practice 4

Complete the missing parts of the program.

```
public class Main {  
    public static void main(String[] args) {  
        _____ subjects = {"Java", "HTML", "CSS"};  
        _____ scores = new int[5];  
        _____ attendance = {true, false, true};  
    }  
}
```